
DIVERGE WITH SEMANTIC STRUCTURE ENHANCEMENT: PRUNING NOISY EDGES FOR TEXT-ATTRIBUTED GRAPH LEARNING

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ABSTRACT

Node classification on text-attributed graphs (TAGs) requires both rich textual semantics and a graph topology aligned with the task. Large language model (LLM) pipelines often improve performance by *rewriting* node text or structure, but such enhancers can be costly, brittle, and dependent on external API access. The **DIVERGE** framework takes a different path: it fine-tunes a pre-trained LLM with LoRA under multiple random initializations to obtain diverse encodings of *native* node text, trains one graph neural network (GNN) per view, and combines their outputs with adaptive ensemble learning rather than uniform logit averaging. We show that even strong LLM representations are limited when observed edges connect semantically dissimilar nodes, violating homophily and degrading message passing.

We extend DIVERGE with a **semantic structure enhancer**—a lightweight, post-hoc module that prunes noisy edges before per-view GNN training, without modifying node text or calling LLMs for topology editing. Two criteria are proposed: (1) *cosine-similarity pruning*, which removes edges whose LLM endpoint embeddings fall below a validation-tuned threshold; and (2) *uncertainty-based pruning*, which removes edges with highest predictive uncertainty from a cross-validated GNN edge predictor (standard deviation or entropy). Each LLM view yields its own refined graph, so the ensemble jointly exploits semantic, structural, and predictive diversity.

Experiments on Cora, CiteSeer, PubMed, and WikiCS show that targeted edge removal consistently improves over DIVERGE alone and outperforms strong baselines including TAPE, while random edge deletion severely hurts accuracy—confirming that gains arise from informed denoising rather than sparsification alone. Cosine and uncertainty enhancers are complementary across datasets, offering an efficient alternative to full LLM enhancer pipelines for TAG learning.

Keywords Text-Attributed Graphs · Large Language Models · Graph Neural Networks · Graph Structure Learning · Ensemble Learning

1 Introduction

Text-attributed graphs (TAGs) are ubiquitous in real-world systems: citation networks link scholarly documents, e-commerce graphs connect products through co-purchase or review relations, and social platforms relate user-generated content through follow or interaction edges [Zhang et al., 2024]. In a TAG, each node is associated with a natural-language description and each edge encodes a relational tie. Node classification—predicting the category of each node from its text and neighborhood—is a fundamental task that requires jointly exploiting *semantic content* and *graph structure*. Graph neural networks (GNNs) excel at relational reasoning through message passing, but their effectiveness depends critically on the quality of node features and on whether the observed topology aligns with the homophily assumption that connected nodes tend to be similar [Kipf and Welling, 2017, Veličković et al., 2018, Jin et al., 2020].

Recent progress has shifted TAG pipelines from shallow text encoders toward large language models (LLMs), which capture richer semantics from node documents [Chen et al., 2024a, Huang et al., 2024, He et al., 2024]. A parallel research line treats LLMs as *enhancers*: they rewrite node text, relabel nodes, synthesize new vertices, or modify graph structure before GNN training [He et al., 2024, Sun et al., 2026, Zhang et al., 2025a, Yu et al., 2025]. Although powerful, enhancer methods often incur API cost, hallucination risk, or multi-stage complexity, and they alter the native textual content of the graph. An alternative strategy asks whether strong performance can arise from *diverse encodings of unchanged node text* rather than from generative augmentation.

DIVERGE addresses this question by fine-tuning a pre-trained LLM with parameter-efficient adaptation (LoRA) under multiple random initializations. Each run produces a distinct semantic view of the same native documents; a separate GNN is trained on each view, and an adaptive ensemble combines their logits—assigning learnable weights at the model or class level rather than uniformly averaging predictions as in TAPE [He et al., 2024]. DIVERGE thus pursues diversity in *how* text is encoded, not in *what* text is provided to the model.

Despite strong semantic representations, a second challenge remains largely unaddressed in this setting: **the observed graph structure may be noisy or misaligned with semantics**. Citation links, hyperlinks, and co-occurrence edges are constructed by rules that do not guarantee semantic similarity between endpoints. When dissimilar nodes are connected, message passing propagates irrelevant information, lowers effective homophily, and limits GNN performance even if node embeddings are high quality [Zhu et al., 2021, Chen et al., 2020, Dong and Kluger, 2023]. Classical graph structure learning (GSL) methods such as ProGNN, IDGL, and PTDNet refine topology using node features [Jin et al., 2020, Chen et al., 2020, Luo et al., 2021], but they typically operate on a single shallow feature space in a joint end-to-end pipeline. LLM-based structural enhancers, meanwhile, rewrite edges or relation types through external model calls [Sun et al., 2026, Guo et al., 2025, Zhang et al., 2025a]. We argue that DIVERGE’s multi-view LLM embeddings offer a natural signal for *lightweight, post-hoc* structure refinement—without generating new text or invoking additional LLM queries at enhancement time.

Our approach. We extend DIVERGE with a **semantic structure enhancer** that prunes noisy edges before per-view GNN training and ensemble aggregation. The key insight is that nodes with similar semantics cluster in LLM embedding space; edges whose endpoints have low cosine similarity are likely to harm homophily-driven message passing. We propose two complementary mechanisms:

- **Cosine-similarity enhancement (Method II).** For each LLM view $\tilde{f}_i$, we compute node embeddings $\mathbf{Z}^{(i)}$ and remove edges (v_i, v_j) whose endpoint representations satisfy $\cos(\mathbf{z}_i, \mathbf{z}_j) < T$ for a threshold T . This yields a collection of view-specific refined graphs $\{\tilde{G}_i\}_{i=1}^M$, each with higher semantic homophily while preserving native text and node features.
- **Uncertainty-based enhancement (Method III).** We train a GNN-based edge predictor with cross-validation and treat edges with high predictive uncertainty—measured by entropy or standard deviation across folds—as unreliable. Removing the most uncertain edges again increases homophily and complements similarity-based pruning with a model-driven criterion related to edge-prediction GSL [Yu et al., 2020, Zhao et al., 2021].

Both enhancers plug into the existing DIVERGE pipeline: multiple LoRA fine-tunes, per-view GNN training on refined graphs, and adaptive ensemble learning over logits. The final model therefore exploits *three* sources of diversity—semantic (repeated LLM encodings), structural (view-specific refined topologies), and predictive (learned ensemble weights).

Contributions. Our main contributions are summarized as follows:

- We identify a structural bottleneck in LLM-on-TAG pipelines: even diverse, high-quality semantic embeddings cannot fully compensate for edges that connect semantically dissimilar nodes, and we motivate homophily-oriented edge pruning in this setting.
- We propose two semantic structure enhancers for DIVERGE—cosine-similarity thresholding and uncertainty-based edge removal—that refine topology using LLM embeddings or GNN edge predictions *without* rewriting node text or calling external LLM APIs for structure modification.
- We integrate structural refinement with DIVERGE’s multi-view encoding and adaptive ensemble framework, producing view-specific denoised graphs rather than a single jointly learned adjacency as in classical GSL [Chen et al., 2020, Jin et al., 2020].
- We conduct extensive experiments on benchmark TAG datasets, showing that removing noisy edges consistently improves both individual GNN accuracy and ensemble performance compared to DIVERGE alone and to representative LLM-enhancer baselines such as TAPE [He et al., 2024].

Paper organization. The remainder of this paper is organized as follows. Section 2 reviews graph structure learning, TAG-specific denoising, and LLM enhancer methods. Section 3 presents notation, a brief DIVERGE recap, and the cosine-similarity and uncertainty-based structural enhancers. Section 4 reports experimental results and ablations. Sections 5 and 6 discuss findings, limitations, and future directions.

2 Related Works

Text-attributed graphs (TAGs) combine relational structure with natural-language node descriptions, making them well suited to citation networks, product graphs, and social platforms where both topology and text carry predictive signal [Zhang et al., 2024]. A central design question is how to fuse textual semantics with graph propagation: early pipelines encoded documents with shallow features (e.g., bag-of-words) and fed the resulting vectors into graph neural networks (GNNs), whereas recent work leverages pre-trained language models (PLMs/LLMs) for richer node representations [Chen et al., 2024a, Huang et al., 2024]. Our work builds on **DIVERGE**, which fine-tunes an LLM with parameter-efficient adaptation (LoRA) under *multiple random initializations* to obtain diverse semantic encodings of *native* node text—without rewriting, paraphrasing, or augmenting the original documents—and then trains one GNN per encoding before combining their outputs with adaptive ensemble learning rather than uniform logit averaging [He et al., 2024]. In this paper we extend DIVERGE with **semantic structure enhancement**: we refine the observed topology when edges connect semantically dissimilar nodes, thereby increasing homophily before message passing. We organize prior work into three lines—general graph structure learning, structure-aware TAG learning without LLM enhancers, and LLM-based enhancement—and then state the research gap our approach addresses.

2.1 Edge Rewiring and Denoising in Graph Learning

Real-world graphs are often noisy, incomplete, or misaligned with node features. Because GNNs propagate information along observed edges, spurious connections can inject irrelevant signals and degrade node classification [Zhu et al., 2021, Jin et al., 2020]. *Graph structure learning* (GSL) addresses this by jointly or sequentially refining topology and GNN parameters.

Joint structure–parameter optimization. **LDS** [Franceschi et al., 2019] formulates structure learning as bilevel optimization over a discrete edge distribution and GCN weights, enabling learning when graphs are missing or corrupted. **ProGNN** [Jin et al., 2020] treats the adjacency matrix as a learnable variable regularized toward sparsity, low rank, and feature smoothness, and has become a standard reference for graph denoising under adversarial or noisy edges. **IDGL** [Chen et al., 2020] iteratively alternates between (i) constructing a new graph from *cosine similarity* of node embeddings and (ii) retraining the GNN on the updated graph until convergence. This iterative, similarity-driven rewiring is particularly relevant to our cosine-based structural enhancer (Method II), although IDGL operates end-to-end on shallow features rather than on post-hoc LLM embeddings from multiple semantic views.

Explicit edge removal and feature-aware rewiring. Several methods target *denoising* rather than full adjacency re-estimation. **PTDNet** [Luo et al., 2021] learns a parameterized edge-drop module trained jointly with the downstream GNN to remove task-irrelevant connections. **GPS** [Dong and Kluger, 2023] introduces an edge signal-to-noise ratio (ESNR) to quantify structural noise and rewires graphs using self-supervised link propensity scores derived from node features. **DIGL** [Klicpera et al., 2019] smooths the adjacency via personalized PageRank diffusion before propagation, offering a classic pre-processing denoising baseline. Together, these works establish that *pruning or rewiring unreliable edges* can substantially improve GNN performance when node features encode meaningful similarity—a motivation we adopt in the TAG setting using LLM-derived semantics.

Edge prediction and graph revision. **GRCN** [Yu et al., 2020] couples a graph-revision GNN that adds or reweights edges with a node-classification GNN optimized jointly on the revised structure. **GAug** [Zhao et al., 2021] uses a graph autoencoder to predict edge existence, adding likely links and dropping unlikely ones before downstream training. These edge-predictor designs parallel the spirit of our uncertainty-based enhancer (Method III), where an encoder–decoder estimates edge reliability; however, GRCN and GAug revise structure inside a single end-to-end pipeline on fixed input features, whereas we use cross-validated prediction uncertainty as a separate, interpretable criterion for noisy-edge removal on graphs already enriched by diverse LLM encodings.

Self-supervised and compact structure learning. **STABLE** [Wei et al., 2022] first learns robust node embeddings via contrastive self-supervision and then rewires the graph by embedding similarity. **CoGSL** [Fatemi et al., 2022] compresses mutual information across multiple graph views to obtain a minimal sufficient structure for the task. **SUBLIME** [Liu et al., 2022] performs unsupervised GSL through structure bootstrapping and contrastive agreement

between learned and anchor graphs. These methods demonstrate that structure can be improved even with limited or no direct supervision on edges, complementing our homophily-driven pruning strategies.

2.2 Edge Denoising for Text-Attributed Graphs

On TAGs, structure quality is equally critical: citation, co-purchase, and hyperlink graphs frequently connect documents that are topologically related yet semantically distant, violating the homophily assumptions underlying standard message passing [Yang et al., 2019, Zhang et al., 2024]. Early textual-network models such as **VHE** [Yang et al., 2019] encode homophily directly in a variational framework, encouraging correlated latent representations for linked nodes while jointly modeling text and topology; however, VHE does not explicitly remove noisy edges from the observed graph. Joint graph-GNN learners including **GLCN** [Jiang et al., 2019] and **GEN** [Wang et al., 2021] estimate or refine adjacency from node features and have been widely evaluated on citation-network benchmarks (Cora, Citeseer, PubMed) with bag-of-words attributes—historically the standard TAG setting before LLM encoders became dominant.

Importantly, most GSL methods above were *not* designed for modern TAG pipelines where node features come from fine-tuned LLMs and where multiple semantic views of the same native text are available. They typically (i) use a single feature source, (ii) entangle structure learning with one GNN training run, and (iii) do not exploit representation diversity from repeated parameter-efficient fine-tuning. Our structural enhancers differ precisely along these axes: we keep native text unchanged, derive *multiple* LLM embedding spaces from DIVERGE, and apply *post-hoc* topology refinement—cosine-threshold pruning (Method II) or uncertainty-based edge removal (Method III)—before training the per-view GNNs and ensemble. This positions our work between classical GSL on citation graphs and recent LLM enhancers that modify text or structure through external model calls (next subsection).

2.3 LLMs as Enhancers for Text-Attributed Graph Learning

A growing line of work assigns LLMs an *enhancer* role: they alter node text, graph structure, or labels *before or during* GNN training to improve downstream performance [Jin et al., 2024, Chen et al., 2024b, Li et al., 2024, Wu et al., 2025]. We group enhancer methods by the modality they modify.

Text and feature enhancement. **TAPE** [He et al., 2024] queries an LLM for zero-shot predictions and natural-language explanations, translates explanations into dense features through an LLM-to-LM interpreter, and trains multiple GNNs whose logits are uniformly averaged. Although highly effective, TAPE depends on external LLM API access and may inject semantic noise when native texts are already informative. **NRUP** [Kuang et al., 2023] constructs a hierarchical text-word graph and pre-trains with self-supervised tasks that co-update text and structure. **E-LLaGNN** [Jaiswal et al., 2024] limits cost by enhancing only a budgeted subset of critical nodes via on-demand LLM prompts, enabling LLM-free inference at deployment. These methods enrich *inputs* to the GNN; DIVERGE instead seeks gains from *diverse encodings of unchanged native text* combined with adaptive ensembling.

Structure and topology enhancement. Structural LLM enhancers modify edges or relation types rather than rewriting node documents. **RoSE** decomposes single-relation edges into multiple semantic relation types using LLM-based generation and classification of edge semantics. **LLM4GraphTopology** [Sun et al., 2026] uses LLM-estimated semantic similarity to add or delete edges and applies pseudo-label propagation as topology regularization during GNN training. **GraphEdit** [Guo et al., 2025] instruction-tunes an LLM to denoise and complete graph structure from a global reasoning perspective. **LoCle** [Zhang et al., 2025a] performs active label-free learning with LLM pseudo-labels and includes GNN-guided topology rewiring under a fixed query budget. Like our work, these methods acknowledge that observed TAG topology may be suboptimal; unlike our work, they rely on LLM rewriting of structure or labels rather than scoring edge reliability from DIVERGE’s multi-view semantic embeddings and lightweight GNN-based uncertainty.

Data generation, label efficiency, and unified pipelines. **LLM4NG** [Yu et al., 2025] synthesizes labeled nodes and predicted edges for few-shot TAG learning. **GAGA** [Xie et al., 2025] annotates a small representative subset of nodes/edges with an LLM and aligns an annotation graph to the full TAG. **UltraTAG** [Zhang et al., 2026] modularizes augmentation, encoding, and GNN training into a unified robust pipeline. **SKETCH** [Zhou et al., 2025] decouples semantic and structural aggregation via retrieval over node texts and distant structural contexts. **TAGA** [Zhang et al., 2025b] builds dual Text-of-Graph and Graph-of-Text views and aligns them through self-supervised learning. Collectively, enhancer methods show that LLMs can improve TAG learning by enriching inputs—often at the cost of API fees, hallucination risk, or complex multi-stage pipelines.

2.4 Research Gap

Existing work leaves three gaps that motivate our DIVERGE extensions with semantic structure enhancement.

Gap 1: Structure refinement without LLM rewriting. GNN-only GSL methods (ProGNN, IDGL, PTDNet, GPS, GRCN, etc.) demonstrate that edge denoising and homophily enhancement benefit message passing, but they generally operate on a *single* shallow feature space and jointly optimize one structure–GNN pair. LLM enhancers improve TAG inputs by *generating* new text, labels, or topology via costly external models. DIVERGE already shows that competitive performance is attainable from *native text alone* when multiple LoRA fine-tunes provide diverse semantic views and adaptive ensembling replaces uniform combination [He et al., 2024]. We ask whether comparable structural gains can be achieved by *refining*—not regenerating—the graph using those same diverse LLM embeddings, without additional LLM API calls at enhancement time.

Gap 2: Multi-view semantics for edge scoring. Prior cosine-similarity GSL (notably IDGL) and feature-aware rewiring (GPS, STABLE) use one embedding source. DIVERGE produces M representation spaces $\{\mathbf{Z}^{(i)}\}_{i=1}^M$ from independently initialized LoRA runs, yielding M structurally refined graphs $\{\tilde{G}_i\}$ rather than a single denoised topology. Method II removes edges whose endpoint embeddings fall below a cosine-similarity threshold in each view; Method III removes edges with highest predictive uncertainty from a cross-validated GNN edge predictor. To our knowledge, combining *multi-view LLM encoding diversity* with *homophily-oriented edge pruning* in a post-hoc enhancer has not been systematically studied for TAG node classification.

Gap 3: Complementarity of similarity-based and uncertainty-based denoising. Cosine pruning (Method II) is interpretable and directly encodes semantic homophily in LLM space, but relies on a fixed similarity threshold. Uncertainty pruning (Method III) is model-driven and captures edges for which relational evidence is ambiguous across data splits, aligning with edge-prediction GSL [Yu et al., 2020, Zhao et al., 2021] yet applied here as a modular step atop DIVERGE rather than as a fully joint rewrite of adjacency during a single GNN training run. Our experiments treat both enhancers as optional modules that plug into the DIVERGE ensemble, allowing semantic representation diversity and structural homophily to be exploited jointly.

Summary of contribution relative to prior work. In contrast to LLM enhancers that augment text or structure through generation [He et al., 2024, Sun et al., 2026, Guo et al., 2025, Zhang et al., 2025a], and in contrast to classical GSL that denoises graphs from shallow features in a single pipeline [Jin et al., 2020, Chen et al., 2020, Luo et al., 2021, Dong and Kluger, 2023], we propose a lightweight **semantic structure enhancer** for DIVERGE: native node text is encoded by multiple fine-tuned LLMs; noisy edges are pruned per view by cosine similarity or predictive uncertainty; GNNs trained on refined graphs feed an adaptive ensemble. This design preserves DIVERGE’s no-augmentation philosophy while importing the homophily motivation of graph rewiring literature into the modern LLM-on-TAG setting.

3 Methodology

3.1 Problem Setup

We study node classification on a text-attributed graph $G = (V, E, D, Y)$, where $V = \{v_1, \dots, v_N\}$ is the node set, $E \subseteq V \times V$ the edge set, $D = \{d_v\}$ the native node texts, and $Y = \{y_v\}$ labels over C classes. The goal is to predict $\hat{y}_v$ for unlabeled nodes using both D and E without modifying the original documents.

3.2 Background: DIVERGE (Brief Recap)

DIVERGE [He et al., 2024] is our base framework. It does *not* augment or rewrite node text; instead it builds diversity from repeated parameter-efficient fine-tuning of one pre-trained LLM $f(\cdot; \mathbf{W})$ with LoRA [Hu et al., 2021]:

$$\bar{f}_i(x) = f(x; \mathbf{W} + \mathbf{B}_i \mathbf{A}_i), \quad (\mathbf{A}_i^{(0)}, \mathbf{B}_i^{(0)}) \sim \mathcal{I}_i, \quad i = 1, \dots, M. \quad (1)$$

Each run i yields node embeddings $\mathbf{Z}^{(i)} \in \mathbb{R}^{N \times p}$ by mean-pooling token hidden states of $\bar{f}_i(d_v)$. A GNN is trained on $(G, \mathbf{Z}^{(i)})$ to produce logits $\mathbf{L}^{(i)}$. Finally, an *adaptive* ensemble (model-, class-, or per-class MLP weighting) combines $\{\mathbf{L}^{(i)}\}_{i=1}^M$ —unlike uniform averaging in TAPE [He et al., 2024]. Algorithm 1 summarizes this pipeline; our contribution inserts a structure-refinement step *before* each per-view GNN.

Algorithm 1 DIVERGE base pipeline (unchanged text; optional structure refinement)**Require:** $G = (V, E, D, Y)$, pre-trained LLM f , LoRA runs M , enhancer $\in \{\emptyset, \text{cosine}, \text{uncertainty}\}$ **Ensure:** Predicted labels $\hat{Y}$ **for** $i = 1$ to M **do** Fine-tune LoRA adapters $\rightarrow$ encoder $\bar{f}_i$ (Eq. 1) Extract embeddings $\mathbf{Z}^{(i)}$ from native texts D **if** enhancer = cosine **then** Build $\bar{G}_i = (V, \bar{E}_i, D, Y)$ by pruning low-similarity edges (Sec. 3.4) **else if** enhancer = uncertainty **then** Build $\bar{G}_i = (V, \bar{E}_i, D, Y)$ by pruning high-uncertainty edges (Sec. 3.5) **else** $\bar{G}_i \leftarrow G$ **end if** Train GNN on $(\bar{G}_i, \mathbf{Z}^{(i)})$; collect logits $\mathbf{L}^{(i)}$ **end for**Adaptive ensemble over $\{\mathbf{L}^{(i)}\} \rightarrow \hat{Y}$ **3.3 Semantic Structure Enhancement**

Motivation. LLM embeddings from DIVERGE encode semantic similarity: linked nodes are often, but not always, close in embedding space. Noisy or rule-induced edges between dissimilar nodes violate homophily and hurt GNN message passing [Jin et al., 2020, Chen et al., 2020]. We therefore refine G into $\bar{G} = (V, \bar{E}, D, Y)$ with $\bar{E} \subseteq E$ by removing unreliable edges *before* training each view-specific GNN. Refinement is applied independently for each embedding $\mathbf{Z}^{(i)}$, yielding $\bar{\mathcal{G}} = \{\bar{G}_i\}_{i=1}^M$ and models $\{\text{GNN}_i(\bar{G}_i, \mathbf{Z}^{(i)})\}_{i=1}^M$. Node texts and features are never altered; only topology changes.

3.4 Method II: Cosine-Similarity Edge Pruning

For view i , let $\mathbf{z}_u^{(i)}$ denote the embedding of node v_u from $\mathbf{Z}^{(i)}$. For each edge $(v_u, v_v) \in E$, define cosine similarity

$$\text{sim}_{uv}^{(i)} = \frac{\mathbf{z}_u^{(i)\top} \mathbf{z}_v^{(i)}}{\|\mathbf{z}_u^{(i)}\|_2 \|\mathbf{z}_v^{(i)}\|_2}. \quad (2)$$

Given threshold $T_i \in [-1, 1]$, the refined edge set is

$$\bar{E}_i = \{(v_u, v_v) \in E : \text{sim}_{uv}^{(i)} \geq T_i\}. \quad (3)$$

Edges with $\text{sim}_{uv}^{(i)} < T_i$ are treated as semantically inconsistent (noisy) and removed. Threshold T_i is tuned on validation data (sensitivity analysis in Section 4). This is a post-hoc, view-specific analogue of similarity-based graph structure learning [Chen et al., 2020] applied to LLM embeddings rather than jointly learned shallow features.

3.5 Method III: Uncertainty-Based Edge Pruning

Cosine pruning uses static embedding distance; uncertainty pruning targets edges whose *existence* is ambiguous under a relational model.

Edge predictor. An encoder GNN maps node features (from $\mathbf{Z}^{(i)}$ or initial GNN states h_u) to representations $\mathbf{z}_u = \text{GNN}(h_u)$. A decoder MLP outputs edge existence probability

$$p_{ij} = \text{MLP}(\mathbf{z}_i, \mathbf{z}_j), \quad (4)$$

trained with binary cross-entropy on observed edges in E .

Uncertainty. The predictor is retrained on M_{cv} bootstrap/cross-validation splits, giving probabilities $\{p_{ij}^{(m)}\}_{m=1}^{M_{cv}}$ per edge. We use either standard deviation

$$U_{ij}^{\text{std}} = \sqrt{\frac{1}{M_{cv}} \sum_m (p_{ij}^{(m)} - \mu_{ij})^2}, \quad \mu_{ij} = \frac{1}{M_{cv}} \sum_m p_{ij}^{(m)}, \quad (5)$$

or entropy $U_{ij}^{\text{ent}} = -\sum_m p_{ij}^{(m)} \log p_{ij}^{(m)}$. High U_{ij} indicates an unreliable edge.

Pruning strategies. Two rules remove noisy edges from E :

- **Top- k %:** remove the k % of edges with highest U_{ij} .
- **Threshold:** remove edges with $U_{ij} > T$.

The result is $\bar{G}_i = (V, \bar{E}_i, D, Y)$ as in Eq. (3). As with Method II, each LLM view i can induce a different $\bar{E}_i$; GNNs and the DIVERGE ensemble proceed unchanged on the refined graphs.

3.6 Overall Pipeline and Complexity

Pipeline. (1) M LoRA fine-tunes on native text $\rightarrow \mathbf{Z}^{(i)}$; (2) optional structure refinement $\bar{G}_i$ via Method II or III; (3) M GNNs on $(\bar{G}_i, \mathbf{Z}^{(i)})$; (4) adaptive ensemble over logits.

Design choices. Methods II and III are modular and mutually exclusive per experiment (or combinable in future work). Neither calls external LLM APIs for structure editing. Method II is simpler and directly encodes semantic homophily; Method III is model-driven and better suited when relational ambiguity dominates [Yu et al., 2020, Zhao et al., 2021].

Cost. Over DIVERGE, Method II adds $O(|E|)$ similarity checks per view; Method III adds M_{cv} edge-predictor training runs—typically small relative to M LoRA fine-tunes.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four standard TAG benchmarks: Cora, CiteSeer, PubMed, and WikiCS [McCallum et al., 2000, Giles et al., 1998, Sen et al., 2008, Mernyei and Cangea, 2022]. Each dataset provides a citation or web-derived graph with text attributes and node-level class labels.

Backbone. We follow the DIVERGE pipeline: a pre-trained LLM is fine-tuned with LoRA under M random initializations (PiSSA, Orthogonal, EVA, Gaussian, LoFTQ), producing M semantic views. Each view feeds a GCN-style GNN trained on the (optionally refined) graph. Three adaptive ensemble variants (Ensemble 1–3) combine per-view logits with learnable weights.

Baselines. We compare against GCN, GraphSAGE, and GAT [Kipf and Welling, 2017, Hamilton et al., 2017, Veličković et al., 2018]; SenBERT and RoBERTa text encoders; ENGINE; TAPE [He et al., 2024]; GraphGPT [Tang et al., 2024]; and LLaGA [Chen et al., 2024a].

Structural enhancers. Method II applies cosine-similarity threshold pruning on LLM embeddings (Sec. 3.4). Method III applies uncertainty-based pruning using standard deviation or entropy from a cross-validated edge predictor (Sec. 3.5). Random removal of 30% of edges serves as a negative control.

Metrics. Node classification accuracy and Macro-F1 on held-out test nodes.

4.2 Main Results

Table 1 compares DIVERGE with cosine-based semantic structure enhancement against strong baselines. Our method achieves the best accuracy on all four datasets, reaching 94.10% on Cora (vs. 90.05% for TAPE) and 97.13% on PubMed. Table 2 reports Macro-F1, where gains are similarly consistent.

The results show that DIVERGE already achieves strong performance without refinement. With cosine-based structural enhancement, further improvements are observed on most datasets, indicating that graph structural enhancement acts as a complementary mechanism for the DIVERGE framework.

4.3 Cosine Similarity Analysis (Method II)

To examine alignment between LLM representations and graph structure, we compute cosine similarity between endpoint embeddings for same-label and different-label edges. Table 3 reports mean and standard deviation across LoRA initialization schemes on Cora.

In all methods, the average cosine similarity for same-label edges is significantly higher than for different-label edges, confirming that LLM representations align with homophily. The higher standard deviation among different-label edges indicates structural noise: some endpoints remain semantically close despite label mismatch. Figure 1 shows clear distributional separation, supporting threshold-based pruning.

Table 1: Node classification accuracy (%): DIVERGE + cosine enhancer vs. baselines.

Method \ Dataset (%)	Cora	CiteSeer	PubMed	WikiCS
GCN	87.41	75.74	89.01	83.67
GraphSAGE	87.44	74.96	90.47	84.86
GAT	86.68	73.73	88.25	83.94
SenBERT-66M	79.61	74.06	94.47	86.51
RoBERTa-355M	83.17	75.90	94.84	87.47
ENGINE	87.00	75.82	90.08	85.44
TAPE	90.05	76.45	93.00	87.11
GraphGPT	82.29	74.67	93.54	82.54
LLaGA	87.55	76.73	90.28	84.03
DIVERGE 1 + Enhancer	93.91	81.03	97.03	89.32
DIVERGE 2 + Enhancer	94.10	81.82	97.11	89.19
DIVERGE 3 + Enhancer	94.10	81.50	97.13	89.28

Table 2: Macro-F1 (%): DIVERGE + cosine enhancer vs. baselines.

Method \ Dataset (%)	Cora	CiteSeer	PubMed	WikiCS
GCN	86.54	71.52	88.54	82.11
GraphSAGE	86.37	71.87	90.16	82.78
GAT	85.64	69.27	87.70	82.14
TAPE	87.21	73.33	92.39	86.03
LLaGA	84.97	72.59	90.00	82.37
DIVERGE 1 + Enhancer	92.65	77.92	96.60	88.43
DIVERGE 2 + Enhancer	92.99	78.56	96.74	88.27
DIVERGE 3 + Enhancer	92.96	78.37	96.77	88.09

Table 3: Mean and std. of cosine similarity for same-label vs. different-label edges (Cora).

Metric	Group	PiSSA	Orthogonal	EVA	Gaussian	LoFTQ
Mean	same_label	0.818	0.903	0.810	0.895	0.799
Mean	diff_label	0.729	0.851	0.724	0.842	0.723
Std	same_label	0.099	0.059	0.097	0.063	0.092
Std	diff_label	0.112	0.080	0.102	0.071	0.096

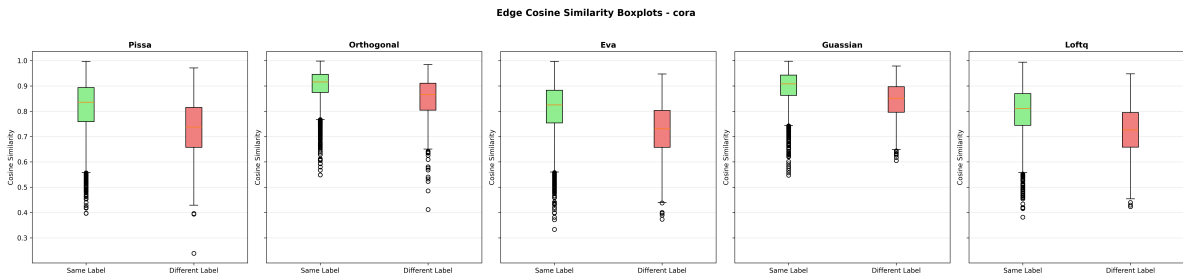


Figure 1: Cosine similarity distributions for same-label vs. different-label edges (Cora).

4.4 Threshold Sensitivity

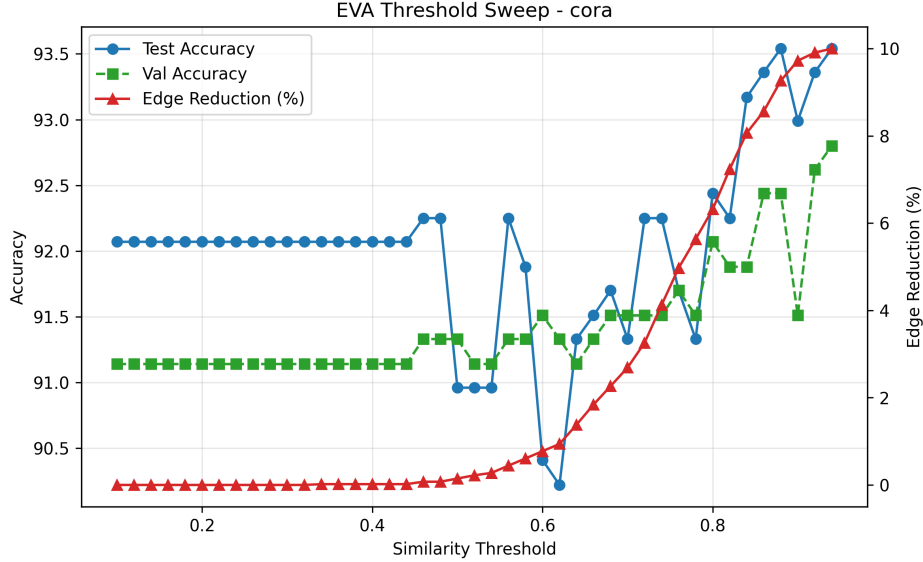


Figure 2: Accuracy and edge-removal rate vs. cosine threshold (representative LoRA init on Cora).

Sensitivity analysis (Figure 2) reveals a practical sweet spot: at low thresholds almost no edges are removed; in a mid-range interval, modest pruning often *increases* accuracy; aggressive pruning eventually deletes useful bridges and hurts performance.

4.5 Comparison of Cosine and Uncertainty Enhancers

Table 4 and Table 5 compare DIVERGE with and without structural refinement across three ensemble variants.

Table 4: Accuracy (%): DIVERGE vs. cosine vs. uncertainty enhancers.

Dataset	Approach	Ens. 1	Ens. 2	Ens. 3
Cora	DIVERGE	93.17	93.36	93.17
	+ Cosine	93.91	94.10	94.10
	+ Uncertainty (std)	93.17	93.17	93.17
	+ Uncertainty (entropy)	92.99	92.80	92.62
PubMed	DIVERGE	96.98	97.13	97.16
	+ Cosine	97.03	97.11	97.13
	+ Uncertainty (std)	97.26	97.16	97.16
	+ Uncertainty (entropy)	97.13	97.11	97.13
CiteSeer	DIVERGE	81.66	81.97	82.13
	+ Cosine	81.03	81.82	81.50
	+ Uncertainty (std)	81.82	82.29	81.66
	+ Uncertainty (entropy)	83.07	82.76	82.92
WikiCS	DIVERGE	88.34	88.38	88.42
	+ Cosine	89.32	89.19	89.28
	+ Uncertainty (std)	88.30	88.38	88.59
	+ Uncertainty (entropy)	88.64	88.51	88.64

Cosine pruning peaks on Cora and WikiCS; uncertainty (std) helps on PubMed; entropy helps on CiteSeer—confirming complementarity between Methods II and III.

Table 5: Macro-F1 (%): DIVERGE vs. enhancers.

Dataset	Approach	Ens. 1	Ens. 2	Ens. 3
Cora	DIVERGE	92.06	91.80	91.85
	+ Cosine	92.65	92.99	92.96
	+ Uncertainty (std)	91.96	91.96	91.96
	+ Uncertainty (entropy)	91.77	91.43	91.59
CiteSeer	DIVERGE	79.25	78.83	79.04
	+ Cosine	77.92	78.56	78.37
	+ Uncertainty (std)	79.25	79.25	79.25
	+ Uncertainty (entropy)	80.72	80.72	80.72

4.6 Random Edge Removal Baseline

Table 6 compares targeted enhancers against random 30% edge deletion on Cora (representative LoRA inits). Random removal causes large accuracy drops, whereas cosine and uncertainty refinement improve or maintain performance.

Table 6: Single-view accuracy (%) on Cora: base vs. enhancers vs. random removal.

Approach	PiSSA	Orthogonal	EVA	Gaussian	LoFTQ
Base	90.77	89.48	92.07	91.14	90.96
Cosine	90.77	90.41	92.07	90.04	91.88
Uncertainty (std)	89.67	91.51	92.44	91.88	91.14
Uncertainty (entropy)	89.48	91.70	92.44	92.25	91.51
Random 30%	88.56	89.59	90.88	90.14	89.41

4.7 Uncertainty Analysis (Method III)

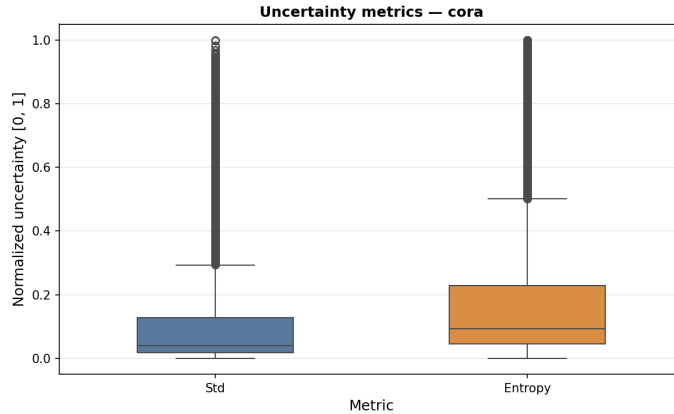


Figure 3: Edge uncertainty distribution on Cora (entropy vs. std).

On Cora, entropy is more dispersed than standard deviation across edges. Increasing the number of cross-validation runs stabilizes estimates with diminishing returns. Std-based removal yields stable PubMed gains; entropy excels on noisier CiteSeer graphs.

5 Discussion

In this section we interpret the experimental findings, relate them to prior work on graph structure learning and LLM-based TAG methods, and discuss limitations and future directions.

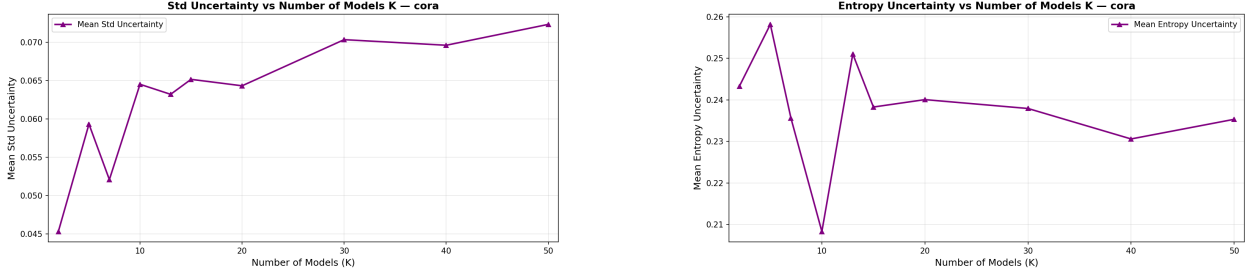


Figure 4: Effect of number of edge-prediction runs on average uncertainty (Cora).

5.1 Semantic Structure Enhancement as a Complement to DIVERGE

Our results support the central hypothesis of this work: *rich LLM semantics and clean graph topology are both necessary* for strong TAG node classification. DIVERGE already achieves competitive or state-of-the-art performance by encoding native node text through multiple LoRA-adapted LLMs and combining per-view GNNs with adaptive ensembling rather than uniform logit averaging [He et al., 2024]. Tables 1 and 4 show that further gains are available when edges connecting semantically dissimilar nodes are removed before message passing.

This pattern aligns with graph structure learning (GSL) literature, which has repeatedly shown that GNNs are sensitive to noisy or task-misaligned edges [Jin et al., 2020, Chen et al., 2020, Dong and Kluger, 2023]. What differs in our setting is the *source* of the feature signal used for denoising. Classical GSL methods such as IDGL and ProGNN refine adjacency from a single shallow or jointly learned embedding space [Chen et al., 2020, Jin et al., 2020]. DIVERGE instead supplies multiple semantic views from independently initialized LoRA fine-tunes; each view induces its own refined graph $\hat{G}_i$, so structural enhancement amplifies representation diversity rather than collapsing structure learning into one end-to-end optimization. The final ensemble thus exploits three complementary axes: **semantic diversity** (multiple LLM encodings), **structural diversity** (view-specific denoised topologies), and **predictive diversity** (learned ensemble weights over GNN logits).

Compared with LLM *enhancer* pipelines that rewrite node text, synthesize labels, or call external models to modify topology [He et al., 2024, Sun et al., 2026, Guo et al., 2025, Zhang et al., 2025a], our structural enhancer is deliberately lightweight: it requires no additional LLM API queries at refinement time and preserves the native document content that DIVERGE was designed to exploit. On Cora, CiteSeer, PubMed, and WikiCS, DIVERGE with cosine-based structural refinement outperforms TAPE, LLaGA, and standard GNN baselines (Tables 1–2), suggesting that homophily-oriented edge pruning can recover part of the benefit otherwise sought through generated explanations or rewritten inputs—while remaining compatible with DIVERGE’s no-augmentation philosophy.

5.2 Why Cosine Similarity Identifies Noisy Edges

The cosine-similarity enhancer (Method II) rests on two empirical observations. First, same-label edges exhibit systematically higher mean cosine similarity than different-label edges in LLM embedding space (Table 3, Figure 1), confirming that fine-tuned LLM representations align with homophily to a useful degree. Second, different-label edges show greater variance, including cases where endpoints remain semantically close despite label mismatch—a signature of structural noise or graph-construction rules that ignore semantic class boundaries.

Threshold-based pruning removes edges whose endpoints are far apart in representation space before GNN propagation. Sensitivity analysis (Figure 2) reveals a practical sweet spot: at low thresholds almost no edges are removed and accuracy is unchanged; in a mid-range interval, a modest fraction of edges is pruned and accuracy often *increases*, indicating that removed connections were harmful rather than informative. Aggressive pruning eventually hurts performance by deleting useful bridges—especially when heterophily or long-range dependencies matter [Dong and Kluger, 2023]. This trade-off parallels PTDNet and GPS, where feature-aware edge removal helps only when sparsification is controlled [Luo et al., 2021, Dong and Kluger, 2023].

Random removal of 30% of edges (Table 6) causes substantial accuracy drops, whereas targeted cosine pruning improves or maintains performance. The contrast confirms that gains are not an artifact of sparsifying the graph per se, but of removing edges specifically inconsistent with LLM-estimated semantic proximity. On Cora, CiteSeer, and WikiCS, cosine refinement yields the largest ensemble gains (Table 4), whereas on PubMed the improvement is smaller—likely because citation links are already relatively homogeneous and the baseline DIVERGE accuracy is already high (~97%).

5.3 Uncertainty-Based Refinement and Edge-Prediction GSL

The uncertainty-based enhancer (Method III) addresses a complementary failure mode. Cosine similarity measures *static semantic distance* in embedding space; it does not explicitly ask whether an edge is supported by relational patterns learnable from the rest of the graph. By training an edge predictor under cross-validation and flagging edges with high standard deviation or entropy across folds, we target connections whose existence is *ambiguous*—a criterion closer to edge-prediction GSL such as GRCN and GAUG [Yu et al., 2020, Zhao et al., 2021].

On Cora (Figure 3), entropy is more sensitive and dispersed than standard deviation. Accordingly, std-based removal tends to yield stable improvements on PubMed (Table 4), while entropy-based removal can excel on CiteSeer. Increasing the number of edge-prediction runs stabilizes uncertainty estimates (Figure 4) but with diminishing returns beyond a modest fold count.

Neither uncertainty measure dominates uniformly. Cosine pruning often achieves larger peak accuracy (e.g., Cora 94.10% vs. 93.36% baseline), while uncertainty pruning can improve Macro-F1 on imbalanced or noisier graphs (e.g., CiteSeer with entropy; PubMed with std in Table 5). This complementarity suggests that semantic distance and relational ambiguity capture partially overlapping but non-identical sets of noisy edges.

5.4 Choosing Between Enhancers in Practice

Based on our experiments, we offer the following practical guidance:

- **Cosine similarity (Method II)** is preferred when LLM embeddings show clear separation between same-label and different-label edges (Table 3), when homophily is a reasonable approximation of the downstream task, and when interpretability matters. It delivers the strongest gains on Cora and WikiCS in our evaluation.
- **Uncertainty with standard deviation (Method III, std)** is preferred when the graph is large, already fairly homophilic, and incremental but stable improvement is desired (PubMed).
- **Uncertainty with entropy (Method III, entropy)** is worth considering on graphs where label noise or heterophily produces many ambiguous links (CiteSeer).
- **DIVERGE multi-view vs. single-view ensemble:** Single-representation ensemble on multiple refined graphs approaches full DIVERGE performance on most datasets, offering a cost-effective deployment option when training multiple LoRA adapters is prohibitive.

A natural extension is to combine criteria, e.g., removing edges that simultaneously fall below a cosine threshold *and* rank among the highest-uncertainty links.

5.5 Relation to Graph Structure Learning and LLM Enhancers

Our semantic structure enhancer occupies a middle ground between GNN-only GSL and LLM-based TAG enhancers (Section 2). Unlike ProGNN, IDGL, and PTDNet, we do not jointly learn a single adjacency matrix inside one training loop; refinement is a modular post-processing step on each LLM view. Unlike RoSE, GraphEdit, and LLM4GraphTopology [Sun et al., 2026, Guo et al., 2025], we do not invoke LLMs to rewrite edges or decompose relation types; structure is inferred from embeddings and GNN edge predictions already computed for classification.

This design trades the expressive power of generative LLM structure editing for *efficiency and reproducibility*: no external API calls, no hallucinated edge labels, and transparent criteria (cosine or uncertainty) for each removed link. The cost is that we cannot add missing edges or discover latent relation types that never appear in the observed graph—capabilities that LLM enhancers explicitly target [Sun et al., 2026, Yu et al., 2025]. For many citation and web-derived TAG benchmarks where the primary issue is *spurious* rather than *missing* edges, pruning-oriented enhancement appears sufficient.

5.6 Limitations

Several limitations should be acknowledged.

Homophily assumption. Both enhancers implicitly assume that removing low-similarity or high-uncertainty edges helps classification. On strongly heterophilic graphs, beneficial edges may connect dissimilar nodes; methods such as ES-GNN explicitly split rather than delete such links [Liu et al., 2024]. Our evaluation focuses on citation-style benchmarks where homophily-oriented refinement is plausible but not guaranteed.

Hyperparameter sensitivity. Cosine threshold T , uncertainty threshold, and top- k removal rate must be tuned per dataset and often per LLM view (Figure 2). Unlike end-to-end GSL, refinement is not automatically aligned with the downstream loss.

Computational overhead. Method II adds pairwise cosine computation over edges for each of M LLM views; Method III adds cross-validated edge-predictor training. Both costs are typically smaller than multiple LoRA fine-tunes but non-negligible on million-edge graphs.

Scope of evaluation. We study node classification on four TAG benchmarks. Extension to link prediction, graph-level tasks, dynamic graphs, or graphs with rich edge text remains open. We also do not compare against every recent LLM enhancer (e.g., UltraTAG, SKETCH, TAGA) [Zhang et al., 2026, Zhou et al., 2025, Zhang et al., 2025b] under identical tuning budgets.

CiteSeer Macro-F1. Cosine refinement improves CiteSeer accuracy but can reduce Macro-F1 relative to baseline DIVERGE (Table 5), suggesting that aggressive pruning may hurt minority classes.

5.7 Future Work

Promising directions include: (i) **adaptive, per-class or per-view thresholds** learned from validation data; (ii) **hybrid enhancers** intersecting cosine and uncertainty scores, or combining pruning with edge addition as in GAUG [Zhao et al., 2021]; (iii) **integration with LLM enhancers** in low-data regimes [Zhang et al., 2025a, Xie et al., 2025]; (iv) **theoretical analysis** connecting removed-edge rate to homophily ratio, building on ESNR and minimal-sufficient-structure frameworks [Dong and Kluger, 2023, Fatemi et al., 2022]; (v) **large-scale and heterophilic benchmarks** (ogbn-products, Roman-Empire); (vi) **single-pass deployment** combining the best refined graphs from one LLM view with multi-graph ensemble learning.

6 Conclusion

We extended the DIVERGE framework for text-attributed graph node classification with a *semantic structure enhancer* that removes noisy edges before GNN message passing, without rewriting native node text or calling external LLMs for topology editing. Two mechanisms—cosine-similarity thresholding on multi-view LLM embeddings and uncertainty-based pruning via cross-validated edge prediction—increase effective homophily and improve ensemble accuracy over DIVERGE alone on Cora, CiteSeer, PubMed, and WikiCS. Targeted removal consistently outperforms random edge deletion, confirming that structure quality mediates the benefit of rich semantic representations. Cosine and uncertainty criteria are complementary: the former excels when embedding-space homophily is strong; the latter offers stable gains when relational ambiguity dominates. By combining native-text LLM diversity with GSL-inspired denoising, our approach offers an efficient alternative to full LLM enhancer pipelines while pushing TAG performance beyond strong baselines including TAPE. Future work will pursue adaptive hybrid enhancers, broader benchmarks, and tighter integration with cost-aware LLM–GNN systems.

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